DATA SOURCING STRATEGY DOCUMENT

Project 1 – Financial Data Analysis | Author: Paul Kinuthia Kuguru | November 2026

1. Purpose and scope

This strategy defines suitable sources for collecting historical financial market data and supporting economic information. The project will focus on daily Open, High, Low, Close and Volume (OHLCV) data for five US-listed companies: AAPL, MSFT, TSLA, AMZN and GOOG. Data will be downloaded as CSV files, retained in the raw-data folder and checked with pandas.

2. Provider comparison

Provider	Data types and coverage	Update frequency	Licensing / access
Yahoo Finance	Equity prices, OHLCV, dividends and splits; broad global instrument coverage.	Daily historical data; intraday availability varies.	Yahoo data terms apply. yfinance is unofficial and intended for personal/research use; verify terms before reuse or redistribution.
Alpha Vantage	Stock OHLCV, adjusted prices, indicators, fundamentals, forex and other markets.	Daily, weekly, monthly and intraday endpoints; freshness depends on entitlement.	API key; free and paid access with endpoint and usage restrictions.
FRED	Economic series such as GDP, inflation, employment, interest rates and exchange rates.	Varies by series: daily, weekly, monthly, quarterly or annual.	Free API subject to terms; some series carry third-party copyright restrictions and require attribution/permission.
Nasdaq Data Link	Financial, economic and alternative datasets from multiple providers.	Varies by dataset and vendor.	Dataset-specific access, fees and licence terms; check the applicable order or dataset terms.

3. AI-assisted evaluation and source selection

The provider comparison was reviewed with AI for relevance, coverage, accessibility and licensing considerations. Yahoo Finance is the practical primary source for this educational OHLCV exercise because it offers convenient historical equity data through Python. Alpha Vantage is the preferred secondary source because it provides documented financial-data APIs and downloadable time series. FRED is suitable for supplementary macroeconomic context, not as the main source for individual-stock OHLCV. Nasdaq Data Link is a useful option when a specific specialist dataset is required and its licence and cost are acceptable.

4. Collection and quality-control procedure

Use yfinance to retrieve daily data for AAPL, MSFT, TSLA, AMZN and GOOG over a consistent historical period. Save each ticker separately in data/raw/ (for example, AAPL.csv). Preserve the original downloaded files. Use pandas to check column names, date order, duplicate dates, missing values, unexpected gaps, invalid OHLC relationships and negative volume. Record the retrieval date, source, ticker, period and any cleaning decisions. Compare selected dates or values with a second provider where practical.

5. Limitations and responsible use

Market data can be delayed, adjusted, incomplete or revised. Trading calendars include weekends and market holidays, so a missing calendar date is not automatically a data defect. API availability and limits may change. The project is for educational analysis, not investment advice. Check the current provider terms before publishing, redistributing or using downloaded data commercially; retain required attribution and respect third-party rights.

6. Reference links

Alpha Vantage documentation: https://www.alphavantage.co/documentation/ | yfinance documentation: https://github.com/ranaroussi/yfinance | FRED API terms: https://fred.stlouisfed.org/docs/api/terms_of_use.html | Nasdaq Data Link terms: https://data.nasdaq.com/terms